

Tanmay Pandey

5th Year BSMS, Indian Institute of Science Education and Research (IISER) Mohali
MS Thesis Internship, Max Planck Institute for Biophysics, Germany

CONTACT

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EDUCATION

2022–2027

BS–MS Dual Degree in Biology
(Physics Minor)
Indian Institute of Science Education and
Research (IISER) Mohali

SKILLS

- **Microscopy:** Fluorescence & phase-contrast; time-lapse imaging
- **Wet-lab:** GPMV isolation; animal cell culture
- **Image Analysis:** CNN segmentation; contour tracking; curvature & fluctuation analysis
- **Computation:** Python, MATLAB; statistics; ML-assisted pipelines
- **Software:** ImageJ (Fiji), Chimera

ADMINISTRATIVE ROLES

2025–2026

General Secretary & Hostel Representative
Student Representative Council (SRC),
IISER Mohali

RESEARCH PROFILE

Final year BS–MS student majoring in Biology with a minor in Physics at IISER Mohali, working on membrane mechanics, plasma membrane vesicles, and quantitative image analysis. Current work involves quantitative analysis of giant plasma membrane vesicles (GPMVs) using computational tools on microscopy data to uncover the physics behind cell death.

RESEARCH EXPERIENCE

Micromechanics of Membrane Blebs in Regulated Necrotic Cell Death

May 2026 – Ongoing

MS Thesis Project – Soft Matter Biophysics Lab, IISER Mohali & Dept. of Membrane Dynamics, MPI Biophysics (Germany)

Co-supervisors: Dr. Uris Ros, Dr. Tripta Bhatia

Investigating membrane mechanical remodeling during regulated cell death using GPMVs, measuring rigidity, tension, permeability, curvature, and shape fluctuations through microscopy and fluctuation spectroscopy. Studying pathway-specific mechanical signatures of cell death using image-analysis pipelines.

Membrane Mechanics and Plasma Membrane Vesicles

Apr 2023 – Apr 2026

Project Intern – Soft Matter Biophysics Lab, IISER Mohali

Supervisor: Dr. Tripta Bhatia

Quantitatively analyzed GPMV and GUV morphology to extract curvature, reduced volume, and membrane mechanical parameters; developed Python pipelines for contour detection, morphological descriptions. Performed GUV electroformation, GPMV preparation, and multi-channel fluorescence microscopy.

DNA-Based Physical Reservoir Computing Systems

Mar 2024 – Dec 2025

Student Assistant (Remote) – Bio-inspired Computation Lab, University of Kiel

Supervisor: Prof. Jan Steinkühler

Developed numerical models of DNA-based spring–mass systems as physical reservoirs inspired by mass-spring dynamics, and analyzed the adaptability and performance of these reservoir networks across nonlinear dynamical tasks.

PUBLICATIONS

- **2026 Quantitative Analysis of Shape Changes in Lipid Vesicles Using Deep Learning-Based Image Segmentation and Tracking Methods** — T. Pandey, R. Kudawla, J. V. B. S., C. Devarakonda, A. Chatterjee, T. Bhatia. *Under revision.*
- **2026 Directed evolution effectively selects for DNA-based physical reservoir computing networks** — T. Pandey, P. Feketa, J. Steinkühler. *Neuromorphic Computing and Engineering (IOP)*. DOI: 10.1088/2634-4386/ae2cc3
- **2025 Experimentally determined shapes of plasma membrane vesicles and PC/PC-cholesterol vesicles** — H. Kaur, R. Kudawla, T. Pandey, T. Bhatia. *Journal of Physical Chemistry B (ACS)*. DOI: 10.1021/acs.jpcb.4c07431
- **2024 Shape analysis of biomimetic and plasma membrane vesicles** — R. Kudawla, H. Kaur, T. Pandey, T. Bhatia. *ChemSystemsChem (Wiley)*. DOI: 10.1002/syst.202400052

RESEARCH SOFTWARE

- **2025 Vesicle Shape Analysis Toolkit** — T. Pandey, T. Bhatia. Open-source software for contour extraction, curvature, and mechanical analysis of vesicle membranes (Zenodo + GitHub). DOI: 10.5281/zenodo.17104430